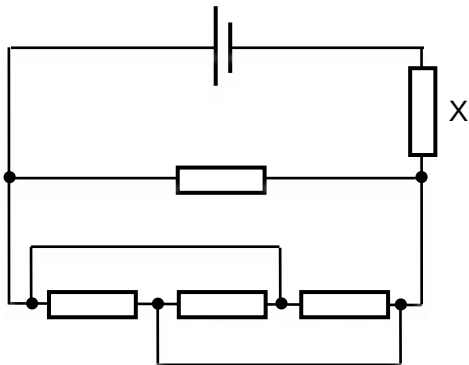


$$I = 60 \text{ A}$$

23 Five identical resistors are connected as shown in the circuit.

If the power dissipated by X is 2.0 W, what is the total power supplied by the battery?



A 2.5 W

B 3.5 W

C 7.5 W

D 10 W

Ans: A

The four resistors that are not labelled are connected in parallel, thus their effective resistance is one quarter that of resistor X, which can be considered to be in series with resistor X. Since power is proportional to resistance for the same current, the power dissipated by the four resistors is going to be $\frac{1}{4}$ that of X. Thus power is 2.5 W.

24 The diagram shows four long straight current-carrying conductors placed at the corners of a